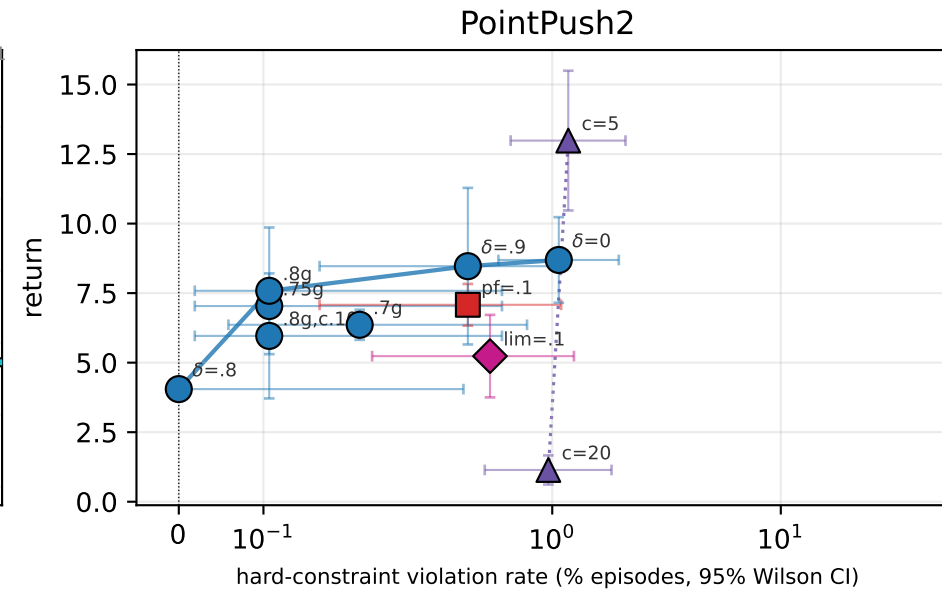
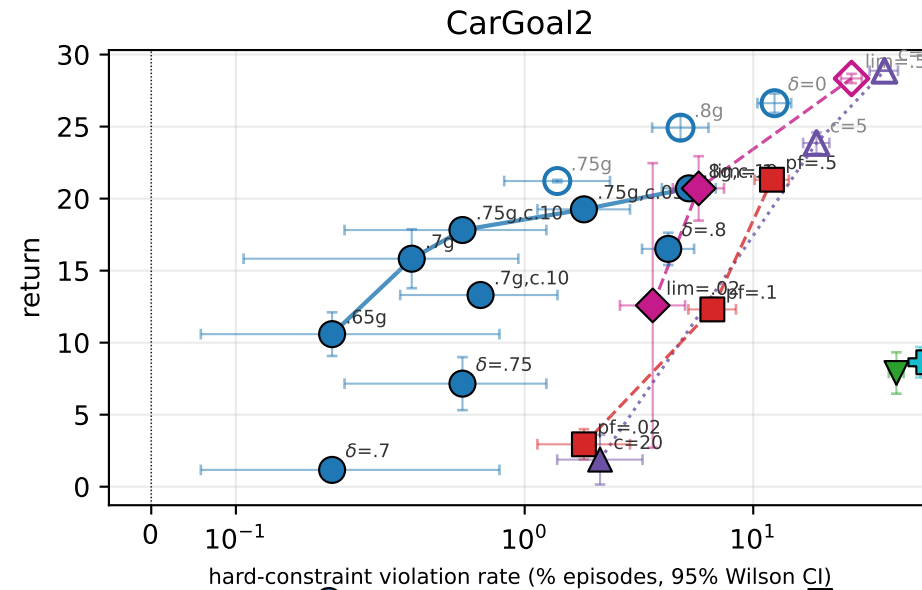
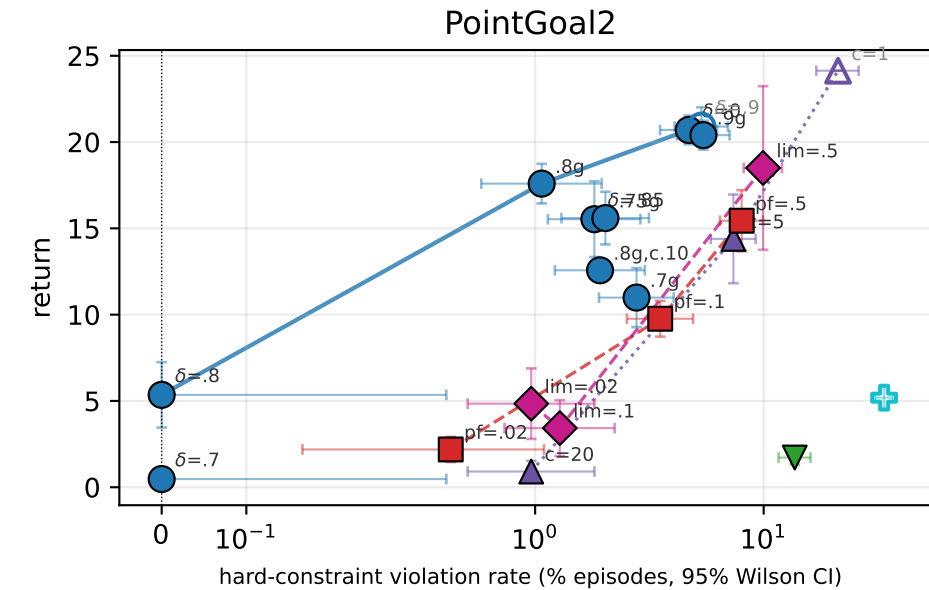


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



SafeOT-typed (ours)

- Safety typed (cars)
- FDR (constraint based)

FDPI (constraint-based)

- ◆ SAC-Lagrangian (penalty-based)

SDAC (multi-constraint SOTA)

SDAC (multi-constraint SOTA)

- ✚ SRCPO (multi-constraint SOTA)

— typed Pareto frontier ($\text{soft} \leq 25$) - - - SAC-Lag cost-limit sweep

hard-constraint violation rate (% episodes, 95% Wilson CI)

..... SAC-Pen coefficient sweep

FDPL is a sugar

--- FDPI p_f sweep

hard-constraint violation rate (% episodes, 95% Wilson CI)

Constraint violation rate (% episodes, 3